

Inverses from Graphs and Tables - Practice

Algebra II Honors | Unit 1 | Lesson 10 Practice | TEK A2.2.B

Objective: I can prove the domain/range swap in general and predict whether an inverse is a function without graphing.

Section 1 — Recall

1. Prove in general: if (a, b) is on f , what point must be on f^{-1} , and why?

Point: _____ Why: _____

2. For f with domain $[2, 9]$ and range $[0, 20]$, state the domain and range of f^{-1} .

Domain of f^{-1} : _____ Range of f^{-1} : _____

Section 2 — Apply

3. A relation $m = \{ (1, 2), (4, 2), (9, 5) \}$. Is m one-to-one? Is m^{-1} a function? Justify using the horizontal-line-test idea, without graphing.

4. A relation $n = \{ (-5, 3), (0, 3), (6, 3) \}$. Is n one-to-one? What does n^{-1} look like, and is it a function?

$n^{-1} = \{ \text{_____} \}$ A function? _____

Section 3 — Challenge

5. Prove algebraically (not by table) that if f is one-to-one, f^{-1} is guaranteed to be a function.

6. A one-to-one function p has $p(3) = 10$, and we are told $p^{-1}(3) = 10$ as well. What must $p(10)$ equal? Explain your reasoning using the swap definition.

 $p(10) = \text{_____}$

Section 4 — Explain It

7. A classmate claims that if $f^{-1}(x) = f(x)$ for even just one specific x -value, then f must be self-inverse everywhere. Is the classmate correct? Explain the difference between agreeing with its inverse at one point versus being self-inverse as a whole.

